

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2012

MA 203: Engineering Mathematics-IV

Date: 14.05.2012, Monday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(a) Find Fourier Sine Transform of the function $f(t) = \frac{e^{-mt}}{t}$, $m > 0$ (4)

(b) Find Fourier Transform of the function $f(t) = \begin{cases} te^{-t}, & \text{for } t > 0 \\ 0, & \text{for } t < 0 \end{cases}$ (3)

Q-2

(a) Find Fourier Cosine Transform of the function $f(x) = e^{-kx^2}$, $k > 0$ (5)

(b) Determine the analytic function $f(z) = u + iv$ whose imaginary part is given by $x^3y - xy^3 + xy + x + y$ (5)

(c) Find the image of circle $|z - 2i| = 2$ under the transformation $w = \frac{1}{z}$ (4)

OR

Q-2

(a) Show that an analytic function with constant modulus is constant. (5)

(b) Prove that the function $f(z) = \sin z$ is an analytic and hence find its derivative. (4)

(c) Define bilinear transformation and find it for which maps the points $(0, -1, i)$ onto the points $(0, 1, \infty)$ (5)

Q-3

(a) By method of Residues, obtain the inverse Laplace transform of $\frac{s+1}{s^2+2s}$ (3)

(b) Evaluate: $\int_C \frac{z^3 - z}{(z-2)^3} dz$, where C is the circle $|z|=3$ (4)

(c) Evaluate: $\int_0^{1+i} (x^2 + iy) dz$ along the path $y = x^2$ (3)

(d) Define Harmonic function; show that $u(x, y) = \frac{1}{2} \log(x^2 + y^2)$ is Harmonic function. (4)

OR**Q-3**

(a) Find the Taylor series expansion for $f(z) = \cos z$ at point $z = \frac{\pi}{4}$ (5)

(b) State Cauchy residue theorem and hence evaluate: $\int_{C|z|=\frac{3}{2}} \frac{4-3z}{z(z-1)(z-2)} dz$ (5)

(c) Obtain Laurent's series expansion of function $f(z) = \frac{1}{z^2 + 4z + 3}$ in the regions $|z| < 1$ and $|z| > 3$ (4)

SECTION-II**Q-4**

(a) Find the Z-transforms of unit impulse function. (3)

(b) Solve the difference equation: $y_{k+1} + \frac{1}{4} y_k = \left(\frac{1}{4}\right)^k$, $k \geq 0$, $y(0) = 0$ (4)

Q-5

(a) Find the inverse Z-transforms of $\frac{4z}{z-a}$ for regions $|z| > |a|$, $|z| < |a|$ (4)

(b) The function $y = f(x)$ is given in the points (7, 3), (8, 1), (9, 1) and (10, 9) using Langrange's interpolation formula. (5)

(c) Compute $\sin(0.54)$ using suitable interpolation formula for the following table: (5)

x	0.5	0.7	0.9	1.1	1.3	1.5
$\sin$	0.47943	0.64422	0.78333	0.89121	0.96356	0.99749

OR

Q-5

(a) Evaluate $\int_0^1 e^{-x^2} dx$ using (i) Trapezoidal rule and (ii) Simpson's $\frac{1}{3^{rd}}$ rule (7)
taking $h = 0.2$

(b) Use Newton-Raphson method to solve the transcendental equation:
 $3x = \cos x + 1$ (7)

Q-6

(a) Find a real root of the equation $x^3 - 3x - 5 = 0$ using Bisection method. (5)

(b) Find by Regula-Falsi method the real root of the equation $\log_{10} x - \cos x = 0$ (5)

(c) Use Taylor's series method; find $y(0.03)$ for $\frac{dy}{dx} = x^2 y - 1$, $y(0) = 1$ (4)

OR

Q-6

(a) Using Picards method, solve the differential equation: $\frac{dy}{dx} = 1 + xy$, $y(0) = 1$ for (5)
 $x = 0.1$

(b) Solve, by Euler's method, the initial value problem $\frac{dy}{dx} = \frac{x-y}{2}$, $y(0) = 1$ over $[0, 3]$,
and using step size $\frac{1}{2}$ (4)

(c) Apply the fourth order Runge-Kutta method to solve:
 $\frac{dy}{dx} = xy$, for $x = 1.2$ initially, $y(1) = 2$. Take step size $h = 0.2$ (5)